ERA MINICON · 16 AUGUST 2026

The ALife Career Landscape

Academia, industry, and the spaces in between — and how to figure out what's right for you.

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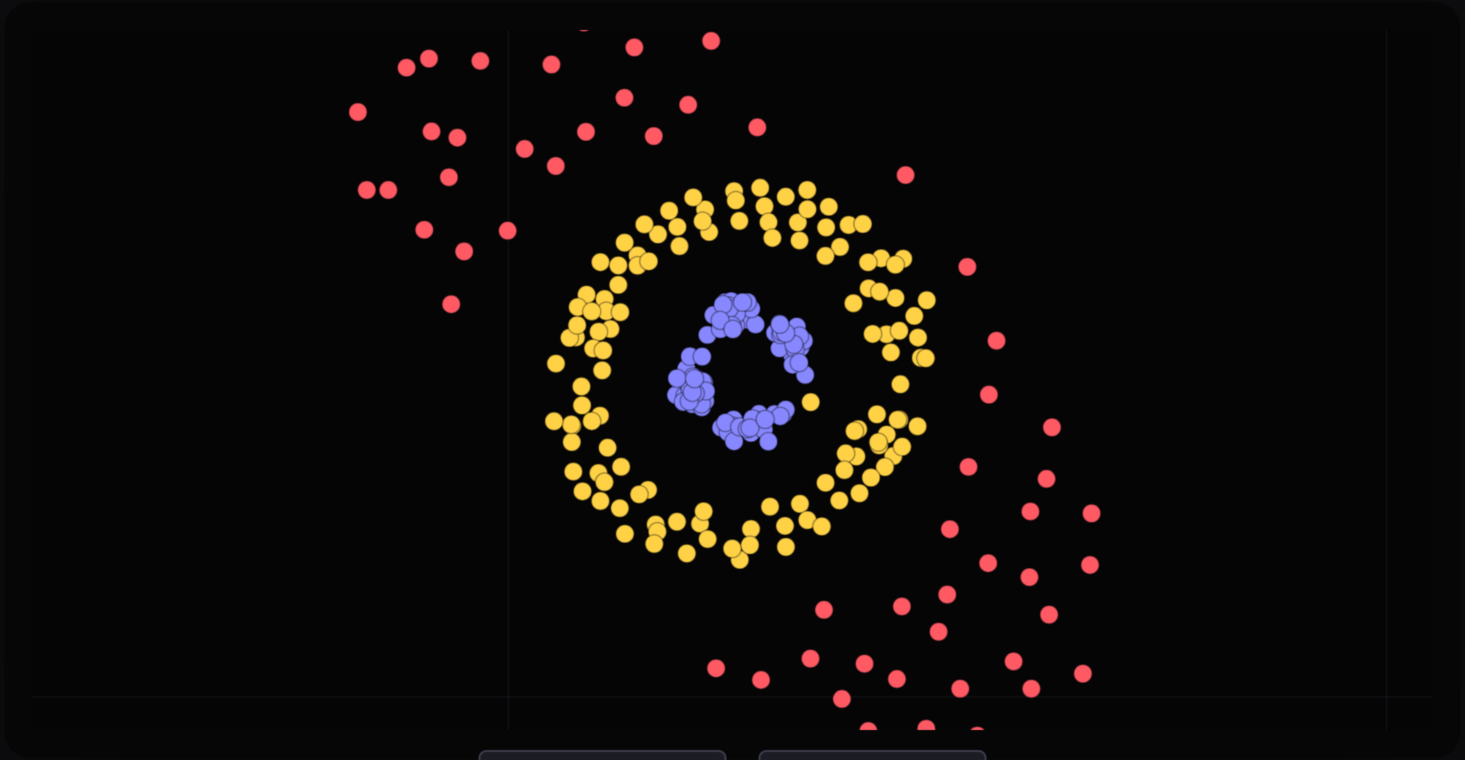
R&D Senior Scientific Research Advisor, Cross Compass
formerly Deputy Director, Cross Labs Kyoto

■ THE FRAME · SWARM CHEMISTRY · SAYAMA 2009

This talk is written in swarm chemistry.

PUBLISHED RECIPE	SPECIES	AGENTS
PULSATING EYE	3	300



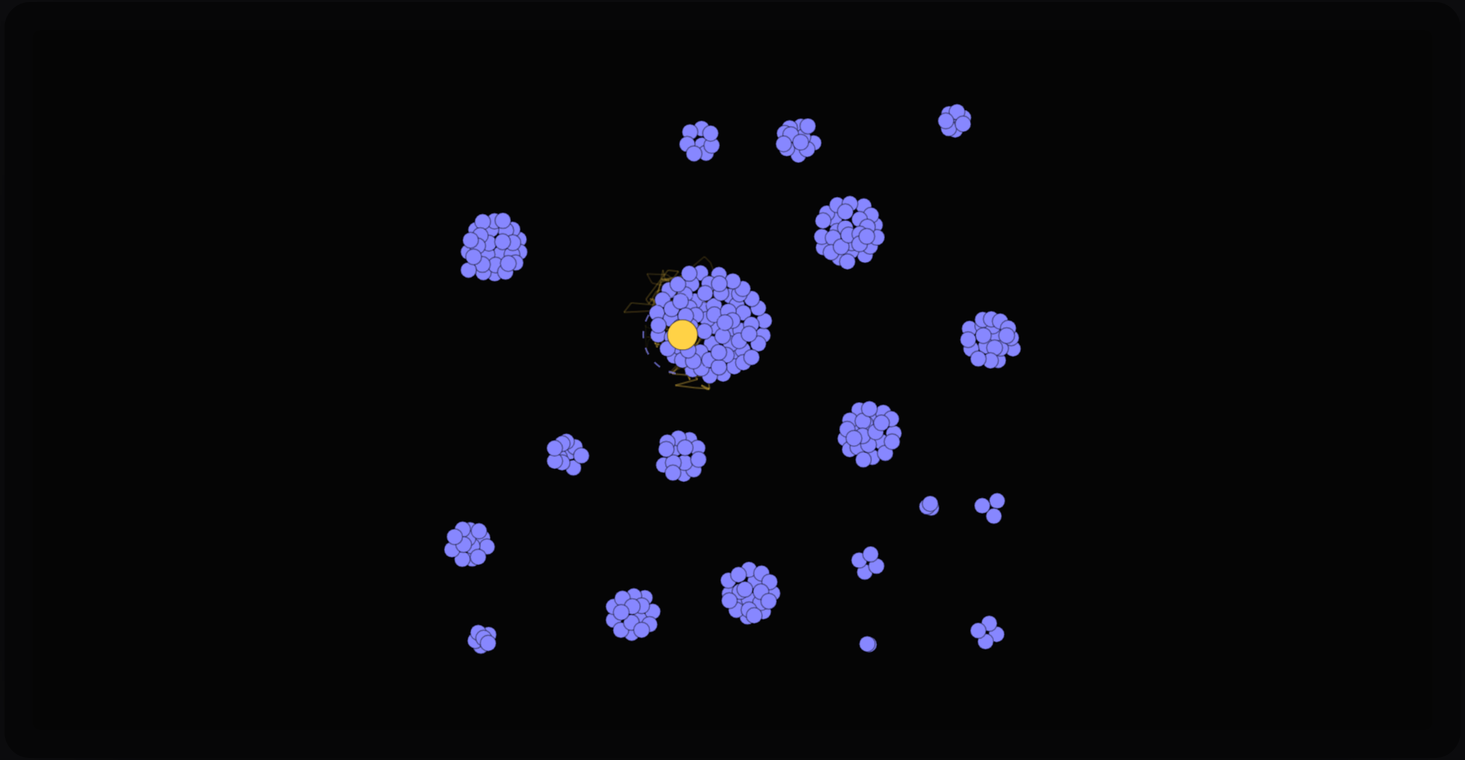
RECIPE ▸ RESET ⟲

Hiroki Sayama's Swarm Chemistry, live — each species senses only inside its own perception radius; cohesion, alignment, separation. Sayama (2009) "Swarm Chemistry," *Artificial Life* 15(1):105–114 · doi · simulator & recipes: bingweb.binghamton.edu/~sayama/SwarmChemistry · parameters unmodified — the names on screen are Sayama's own

■ THE FRAME · ONE AGENT, UP CLOSE

One agent. Two kinds of sensors.

OUTWARD · R	INWARD · GAIN	SENSES	SPEED
20.8	0.68	30	9.8



OUTWARD REACH blind everyone INWARD GAIN numb insistent RESET ⟲

Same physics, camera on one agent. Outward sensors: perception radius, response to neighbours. Inward: its own pace, its own noise. Drag either and the same swarm hands it a different life. · these two sliders are the talk

Artificial life is a method.

> *"...an interdisciplinary study of life and life-like processes that uses a synthetic methodology."*
>
> Mark A. Bedau, *Trends in Cognitive Sciences* 7(11), 2003

DESIGNING YOUR LIFE · THE FIRST INSTRUMENT

A career is a subsystem of a life.

TWO GUIDES

Both books run on serendipity.

TREASURE HUNTER

STANLEY & LEHMAN · 2015
CH. 4: "THE FALSE COMPASS"

COMPASS

BURNETT & EVANS · 2016
THEY CALL IT "WAYFINDING"

Stanley & Lehman, *Why Greatness Cannot Be Planned*, Springer 2015 · doi.org/10.1007/978-3-319-15524-1 · Burnett & Evans, *Designing Your Life*, Knopf 2016, ISBN 978-1-101-87532-2 · Stanley telling his own career non-objectively: *Cross Roads #42*, Cross Labs AI, 14 Mar 2024 — I hosted it · youtube.com/watch?v=73svHfR3eo8 · "the alternative to the objective path is the path of the interesting" (26:46) · quote is from auto-captions — check it against the audio before speaking it

PICBREEDER

One user's alien face became another user's car.

SKULL · 74 GENS

BUTTERFLY · 104 GENS

ALIEN FACE · **ACAMPBEL**

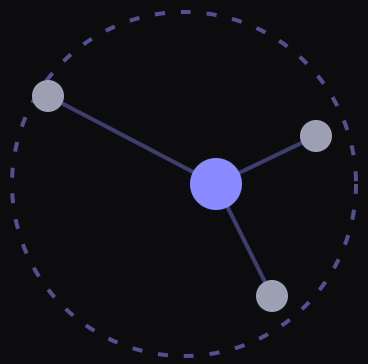
CAR · **KEN (STANLEY)**

"I was not trying to breed a car." — Ken Stanley

Real specimens, rendered from the original CPPN genomes (picbreeder.net), verified against the archived site (pixel r ≥ 0.98) and Secretan et al. (2011) fig. 13, *Evolutionary Computation* 19(3):373–403 · doi · quote: UCF faculty feature · two different users, per the site's own record: acampbel published the face; Stanley branched it — the eyes became the wheels. Skull = the objective-search-fails specimen (Woolley & Stanley 2011); car = the serendipity one.

SECTION 1

Two kinds of sensing.



EXTEROCEPTION

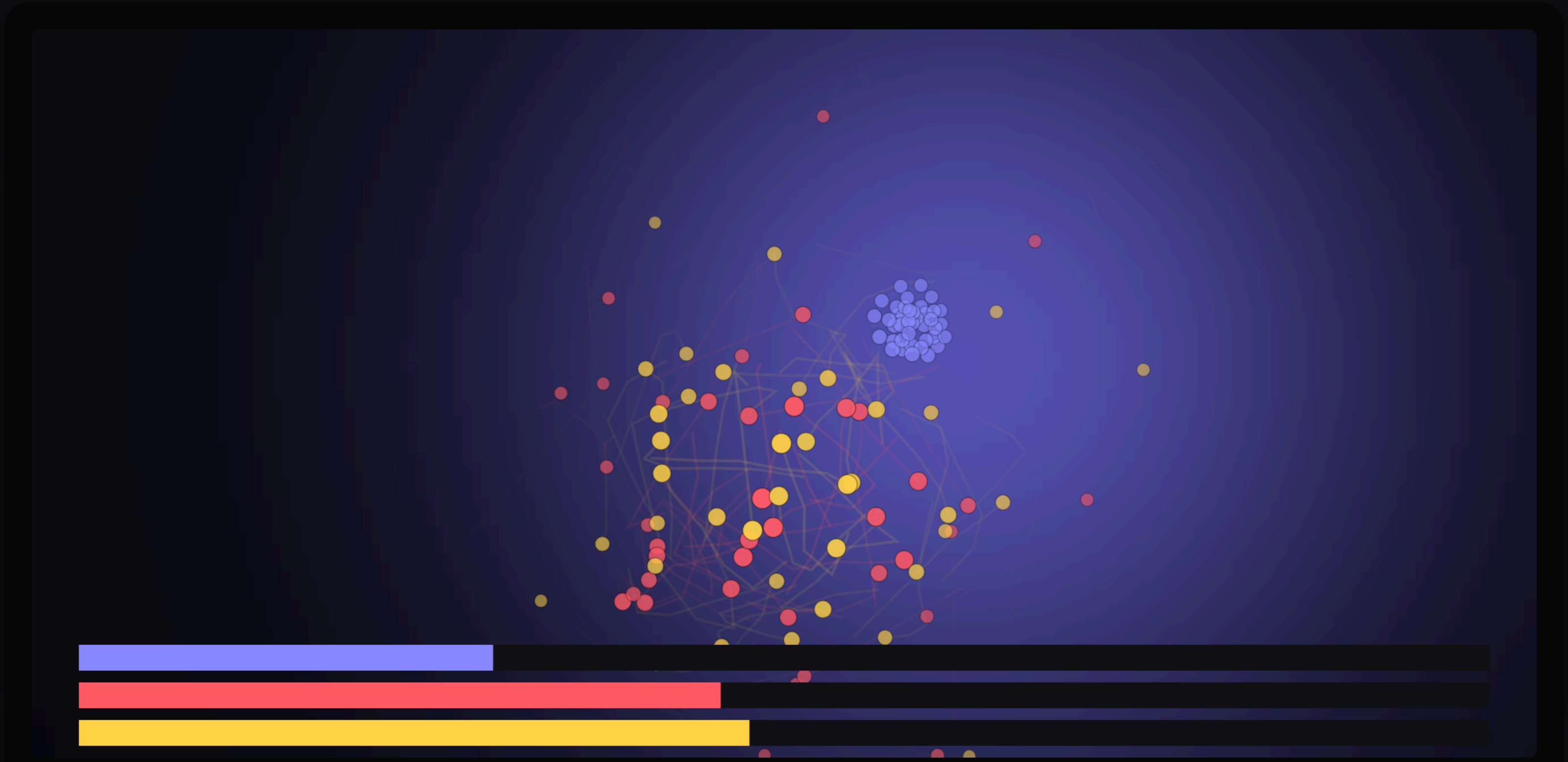
THE OTHERS · A VECTOR

INTEROCEPTION

YOURSELF · A SCALAR

■ SECTION 1 · THE VISIBLE FIELD AND THE TRUE ONE

Everything they can see says they're winning.



● WORLD-ONLY ● SELF-ONLY, SEES NO ONE ● BOTH · BOTTOM BARS = AVG FELT ENERGY PER FLOCK

WORLD FIELDS apart aligned GOLD INNER SENSE none only TRUE FIELD RESET ⟲

Boids: Reynolds (1987) "Flocks, Herds, and Schools," *SIGGRAPH* 21(4):25–34 · doi — flocking is exteroception here, so the red interoception-only population cannot flock · run-and-tumble chemotaxis as the inner mechanism · interoception vs exteroception: Sherrington (1906).

THE SCOPE CONDITION

Stanley draws the line himself.

GETTING LUNCH

"You can make that an objective and that's very reasonable."

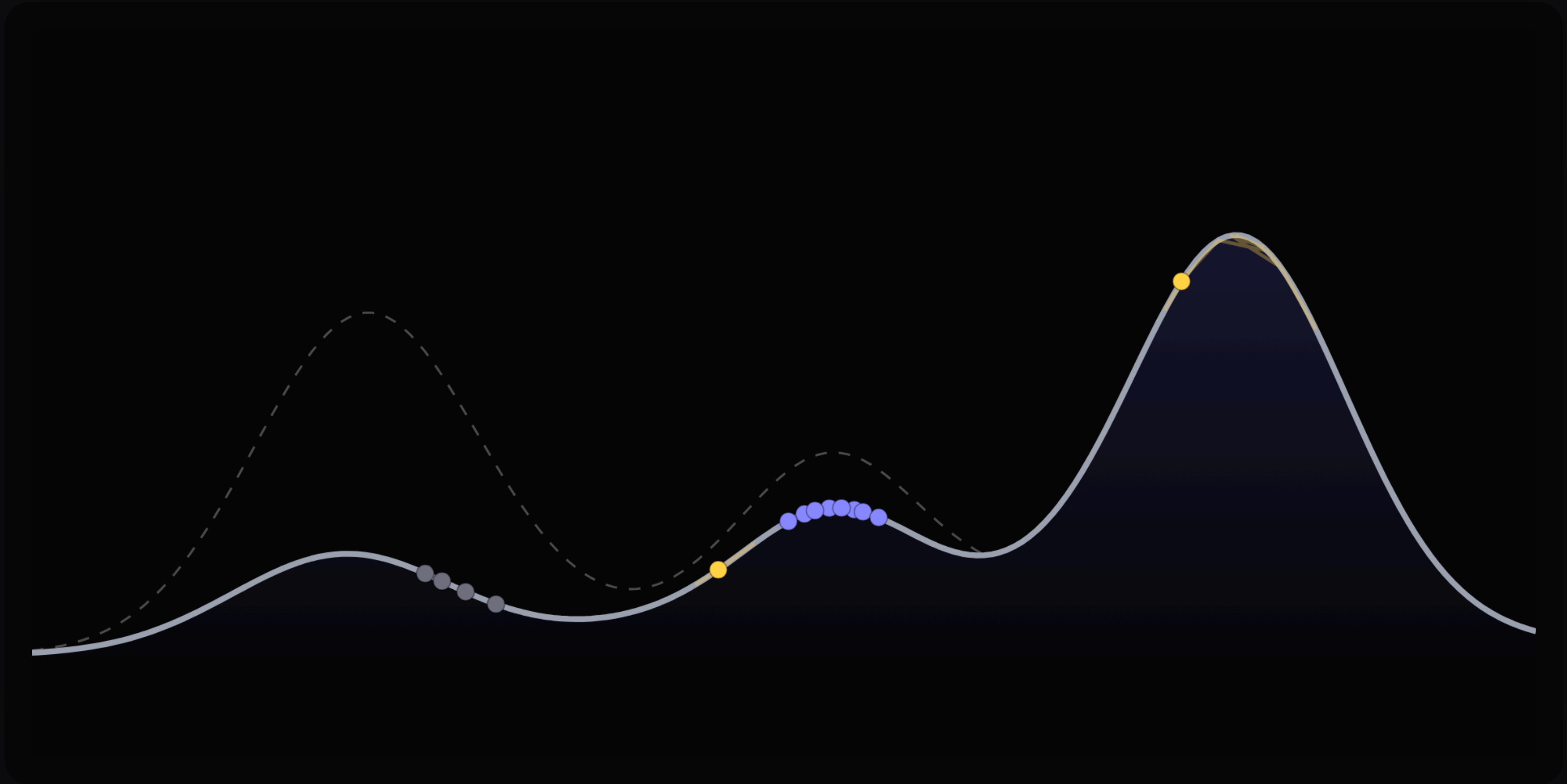
CURING CANCER · CREATING AI

"We just don't know what the intervening stepping stones are... there clearly we have no idea where we have to travel."

KEN STANLEY · THE JIM RUTT SHOW EP130 · 2021

■ SECTION 2 · THE ENVIRONMENT IS NON-STATIONARY

The ground is moving — partly because you moved.



● CLIMBERS ● EXPLORERS ● STRANDED — OUT OF FUEL, CAN REVIVE --- GROUND AT START · X WHAT YOU WORK ON · Y HOW WELL IT GOES

CHANGE RATE SLOW FAST CROWDING OFF STRONG RESET ⟲

Moving-peaks-style non-stationary optimisation: Branke (1999) *CEC* · doi.org/10.1109/CEC.1999.785502 · crowding term after McPherson & Ranger-Moore, "Evolution on a Dancing Landscape," *Social Forces* 70(1), 1991 · doi.org/10.2307/2580060

Nobody made a mistake.

Every stranded agent is still at a local maximum.

The peak really did move.

GOOGLE DEEPMIND · 2024

Position paper: open-endedness is *essential* for superhuman AI.

SAKANA AI · 2026

Three artificial-life releases in four months — one cites *the book* by name.

Hughes et al. (2024) arXiv:2406.04268 · arxiv.org/abs/2406.04268 — cites the journal *Artificial Life* four times and concedes ALife's subjectivity critique by name · sakana.ai/digital-ecosystem · smart-cellular-bricks · picbreeder-ai · Stanley's own latest is an *ALife* paper: arXiv:2607.02954, 3 Jul 2026 · US funding: **proposed** −56.9% vs **enacted** −3.4%; GRFP 2,600→1,500→2,500, second-years excluded; 29% of non-US physics PhDs left within six months (AIP). Full table: backup 39. · proposed, enacted and enjoined are three different things

Designing Your Life is MAP-Elites.

QUALITY-DIVERSITY	DESIGNING YOUR LIFE
Local quality criterion	Workview / Lifeview
Behaviour characterisation	The dimensions you vary a life along
Archive of diverse elites	Three Odyssey Plans, unranked
Cheap evaluation	Prototype conversations

Lehman & Stanley (2011) *Evol. Comput.* 19(2) · doi — "make novelty an objective and fitness another objective in a multi-objective formulation," inside "Abandoning Objectives" itself: QD factorised the compass from heading into local refinement · Mouret & Clune (2015) MAP-Elites · arXiv:1504.04909 · Pugh, Soros & Stanley (2016) *Front. Robot. AI* 3:40 · doi

WORKSHEET P1 · Q2 NOW

▪ YOUR TURN · WE DESIGN THIS INSTRUMENT TOGETHER

Your reward function — inferred from what you actually did.

1 · THE TRACE

Which slice of recent behaviour do we sample?

2 · THE AXES

What two scores does every item get?

3 · THE PROBE

One question we ask of the worst row.

In chat: propose · 👍 what you like · I paste the winning design, then the timer runs.

2:15

■ AEIOU · YOUR HIGHEST-ENGAGEMENT ACTIVITY

A Activities

E Environments

I Interactions

O Objects

U Users

2:30

AEIOU is not a DYL invention — it is an ethnographic field-observation framework from design research (Doblin Group, c. 1991), repurposed from observing other people to observing yourself.

Now find your lowest-energy activity.

Who chose it?

In chat: one word

You just did inverse RL on yourself.

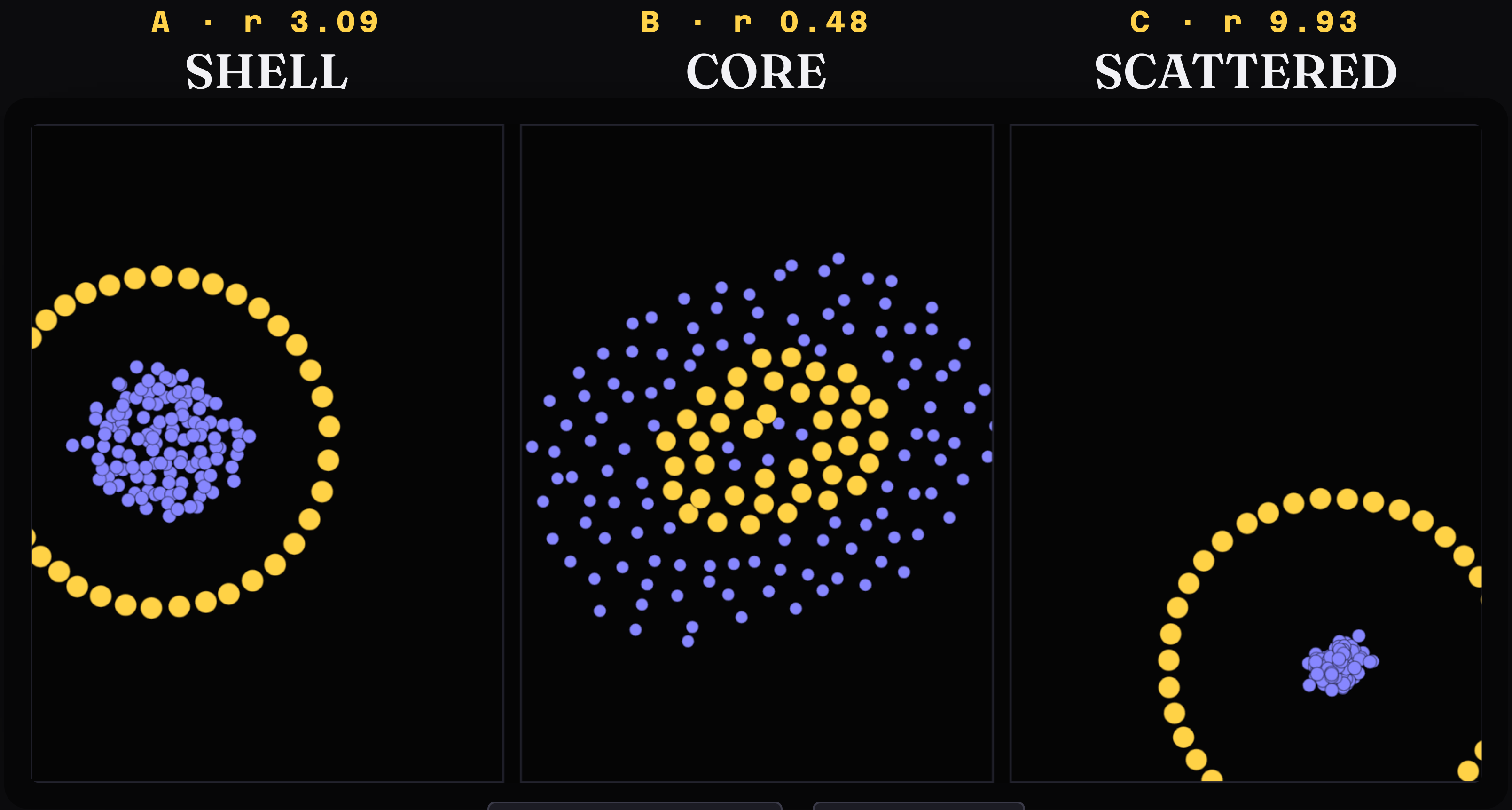
RL	REWARD	→	BEHAVIOUR
IRL	BEHAVIOUR	→	REWARD

SECTION 3

You are in the wrong environment.

▪ SECTION 3 · THE SAME AGENT, THREE WORLDS

Identical recipe. Three mixtures.



RESHUFFLE ⟳ RESET ⟲

Same 40 gold agents in every pane — one published Sayama ingredient, unmodified (Cell with Two Nuclei, 4/6). Ambient mixtures are published ingredients too: Cell 5/6 (A) · Pulsating Eye 3/3 (B) · Pulsating Eye 1/3 (C). Sayama (2009) *Artificial Life* 15(1) · doi · recipes · the pairings are ours, unanimous over 16 seeds each — RESHUFFLE reruns it live

22 EMPLOYERS · EVERY LIVE POSTING · 2026-08-07

2,400 jobs. Zero.

open-endedness · emergence · evolution · self-organisation · artificial life · quality-diversity · novelty · swarm · complex systems

JOBS.AC.UK, SAME DAY: MACHINE LEARNING 142 · ARTIFICIAL LIFE 0

Greenhouse / Lever / Ashby job APIs, 22 employers, and jobs.ac.uk — all fetched 2026-08-07. One title matched — school curriculum, a false positive. The 142 is the control. · re-run the morning of the talk if you want the numbers exact

One job title says "Open-endedness."

SVP, Chemistry	Lila Sciences
SVP, Open-endedness — Kenneth Stanley	Lila Sciences
122 open roles — none of them say it	

lila.ai/team/kenneth-stanley · kenstanley.net · both verified 2026-08-07 — the title was written for the person, not the role

What the postings call it.

YOUR PHRASING	THEIR PHRASING
Evaluating with no ground truth	**evals** — 19 live titles
Open-ended environment generation	**environments** — 8 live titles
Collective behaviour, multi-agent dynamics	**agents** · orchestration
Whole-cell simulation	**"Virtual Cell"** — actual postdocs

The gap is craft, not concepts.

Verbatim, live 2026-08-07: "Research Engineer, Frontier Evals & Environments" (OpenAI) · "Staff Software Engineer, Environments Infrastructure" (Anthropic) · "Member of Technical Staff, Evaluation Execution" (METR) · Virtual Cell postdocs: Arc Institute · the honest arithmetic: ~13 of 2,400 are research-track AND ALife-adjacent — about a dozen openings worldwide · barriers, both real: Anthropic RE, evals ($500–850k, asks for a *bachelor's*; a listed project: "take a flaky distributed eval pipeline and make it boring") · Apollo Research ("we don't require a formal background") · closer to home: Cross Labs · Sakana AI · Sony CSL Kyoto — which does list 人工生命

▪ YOUR TURN · 6 MINUTES · WE READ THESE OUT

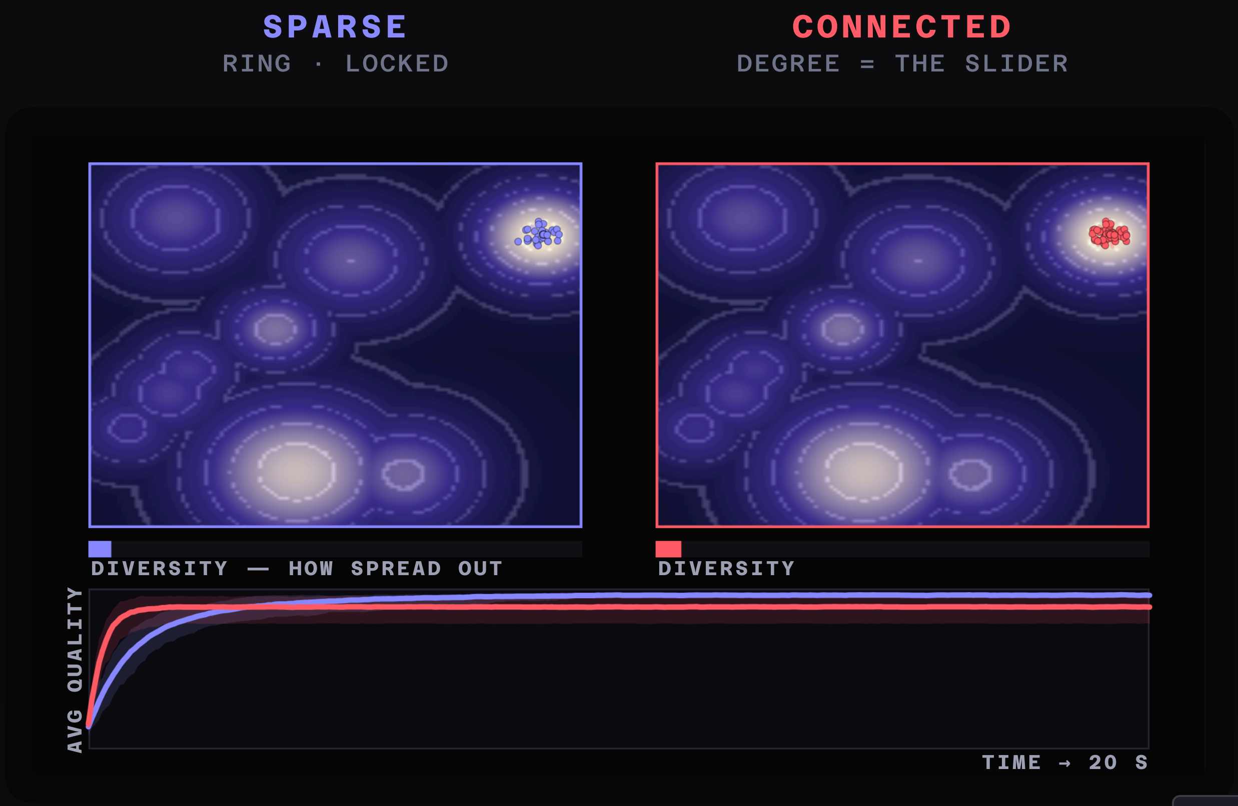
```
I build / study ______
so that ______
can ______
```

It must survive a hiring manager repeating it to their boss.

3:00

One difference: who can see whom.

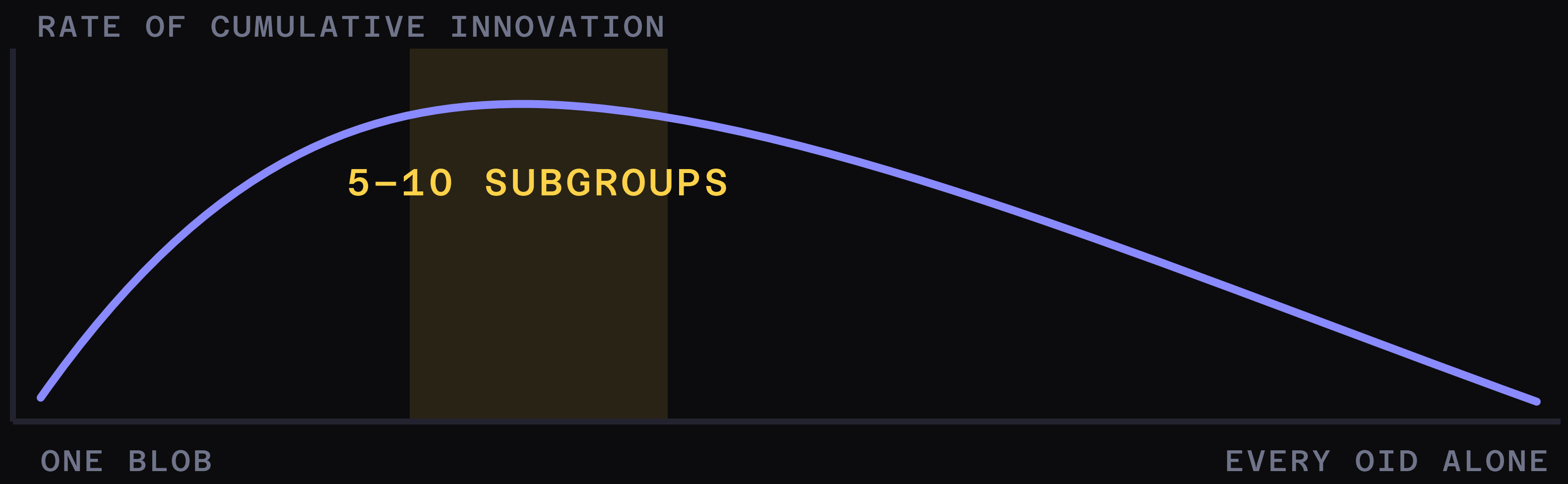
You just made your method legible. Here's a community where everyone can read everyone's.



CONNECTIVITY (RIGHT PANEL) SPARSE EVERYONE RESET ⟲

Lazer & Friedman (2007), *Administrative Science Quarterly* 52(4):667–694 · doi.org/10.2189/asqu.52.4.667 · rule verbatim from the authors' own poster: tortoisehare_poster.pdf · lines are the mean of 16 replicates; bands are the full spread

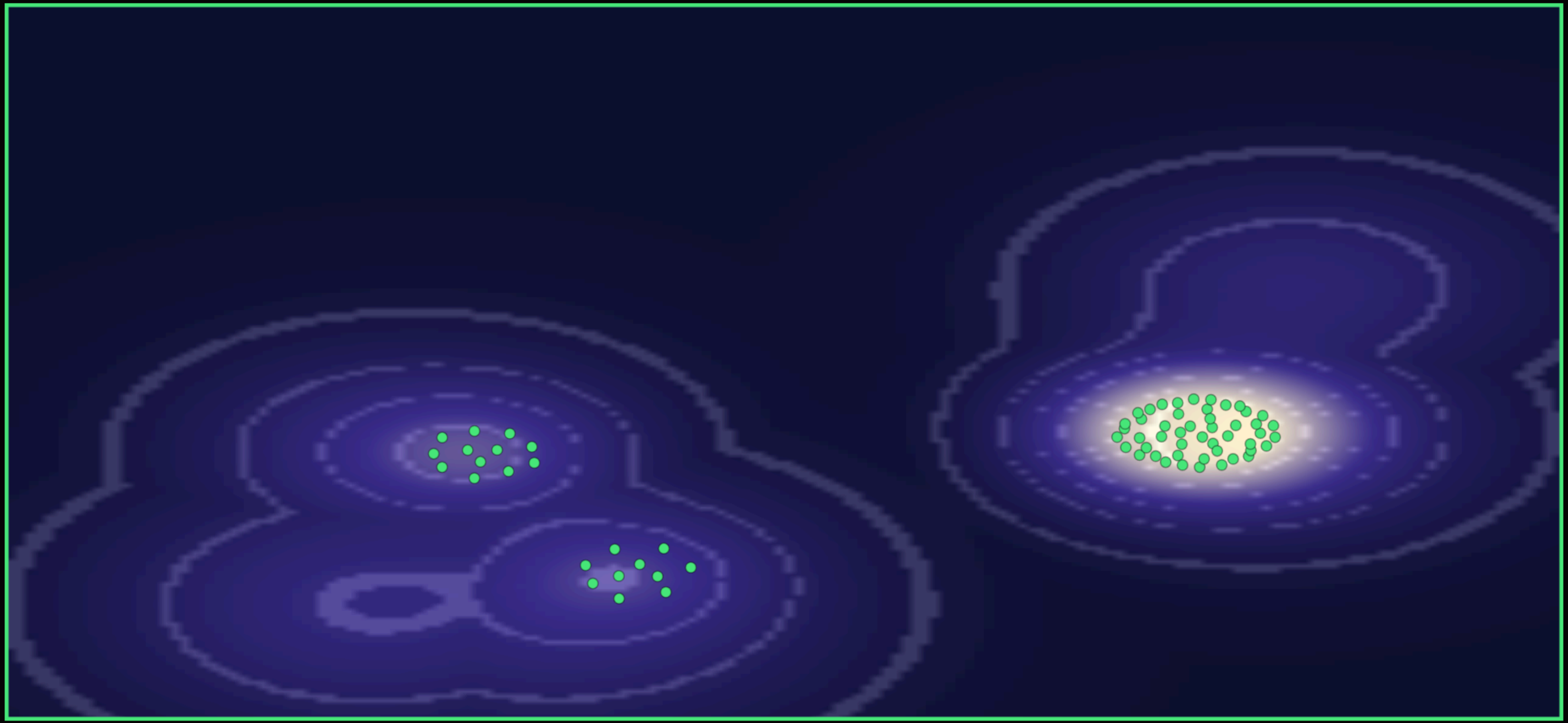
The optimum is 5 to 10 clusters.



curve: Derex, Perreault & Boyd (2018) *Phil. Trans. R. Soc. B* 373:20170062, 600 people · doi — connected groups "maintain complex cultural traits but produce insufficient variation"; fragmented ones the reverse · and it blinks: Bernstein, Shore & Lazer (2018) *PNAS* 115(35) · doi — triads seeing each other constantly / every third round / never found the optimum 33.3% / **48.3%** / 44.1% — separate experiment, separate axis; full table: backup · the copying result reverses under conformity (Barkoczi & Galesic 2016) — say this if asked

SECTION 4 · NICHE CONSTRUCTION

The ground rises where they stand.



SCATTER THEM RESET ⟲

Spatial niche-construction models: Silver & Di Paolo (2006) *Theor. Pop. Biol.* 70:387–400 · Laland, Odling-Smee & Feldman (1999) *PNAS* 96:10242–10247 · doi · applied to a Discord server this is frankly a metaphor — but protected niches in innovation studies, and the sociology of scientific movements, are not

■ YOUR TURN · 5 MINUTES

What niche does this community need that doesn't exist yet?

an industry directory · mentor matching · a "what I actually do all day" series · a CV-translation clinic · a job board that isn't empty

> Post it in chat. Anything with traction, **I take to the ISAL board.**

4:00

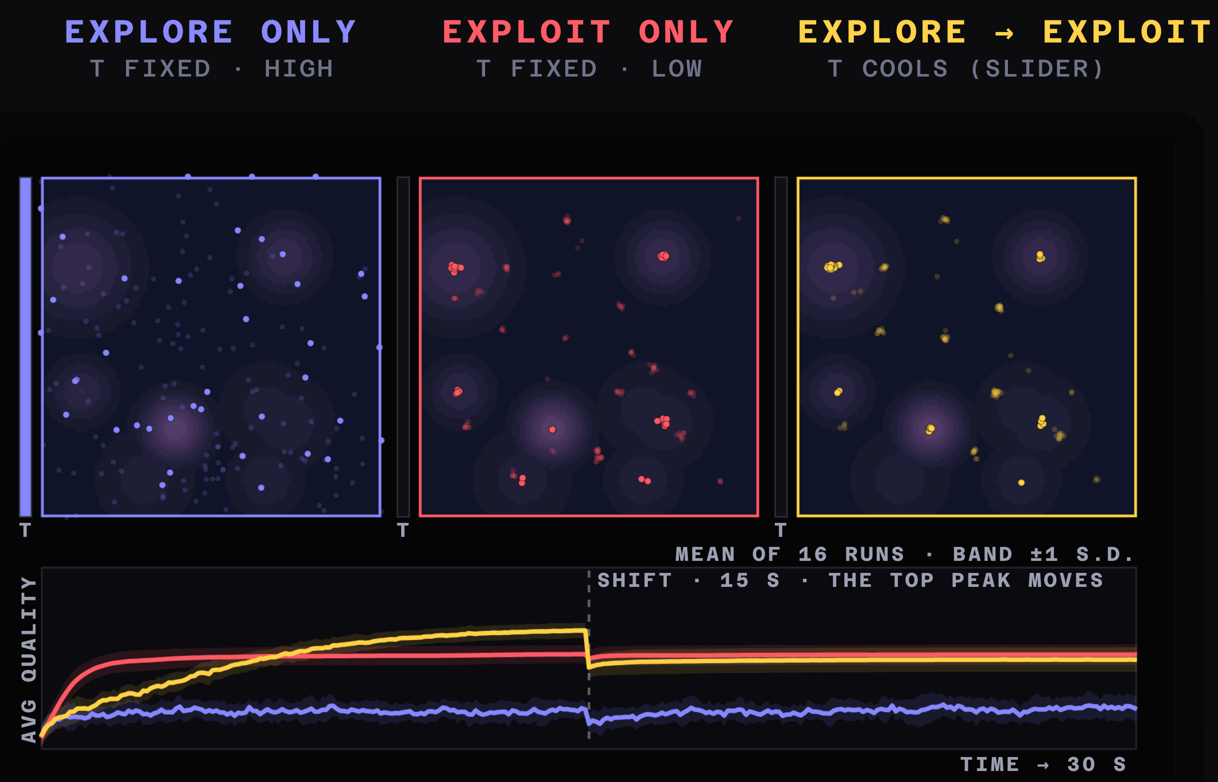
550

of you raised the ground under this community.

ERA Minicon 2026 announcement — "550+ member Discord" · the count is the community's own

The ground moves at fifteen seconds.

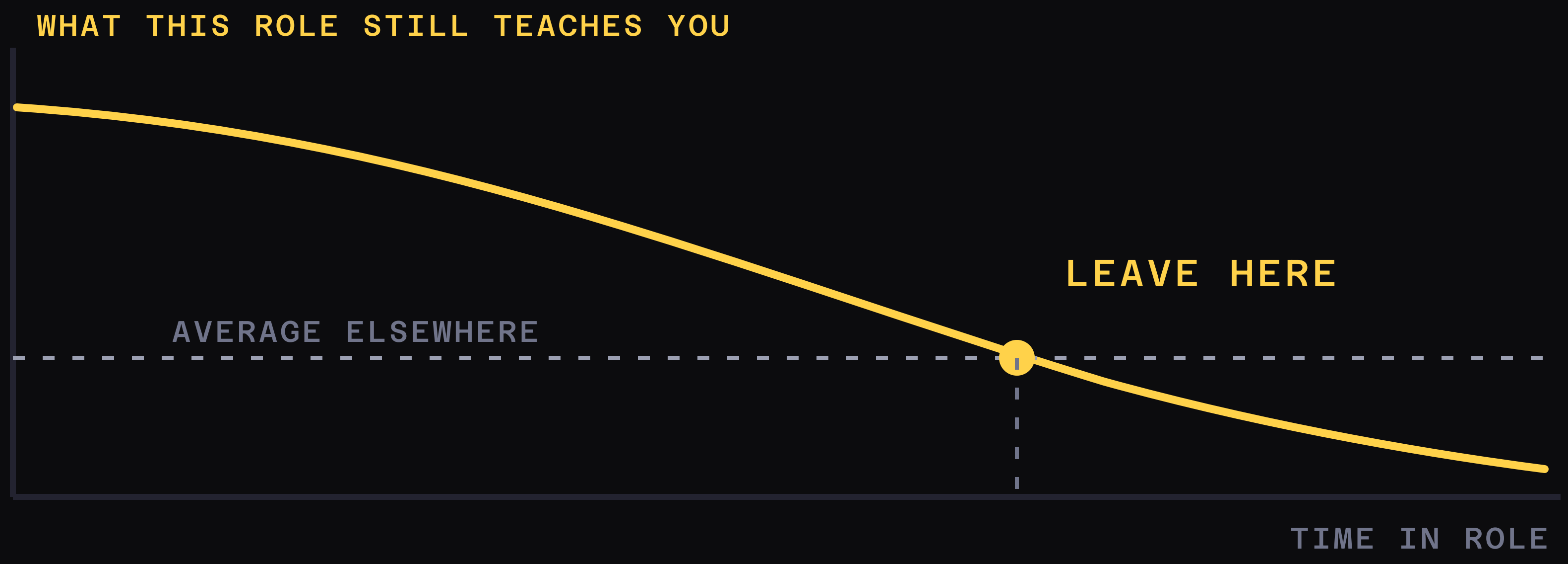
Raising the ground was the community's move. When to commit is only yours.



COOLING (GOLD PANEL) FAST SLOW PIVOT RULE RESET ⟲

Metropolis acceptance = simulated annealing: Kirkpatrick, Gelatt & Vecchi (1983) *Science* 220(4598):671–680 · doi · lines are the mean of 16 replicates, bands ±1 s.d. · press PIVOT RULE for the fourth strategy

Leave when your learning rate crosses the average.



Charnov (1976) *Theoretical Population Biology* 9(2):129–136 · doi — the theorem needs something to deplete; salary doesn't, learning does. Applied to a non-depleting patch it says "never leave." · already an algorithm: Oudeyer, Kaplan & Hafner (2007), Intelligent Adaptive Curiosity, rewards *learning progress* · inria.hal.science

≈26,000 CAREERS · ARTISTS, FILM DIRECTORS, SCIENTISTS

Explore first. Then commit.

A hot streak is the four-or-so-year run when your work lands far above your own average. Nine careers in ten contain one; your output *rate* doesn't change during it; it can start at any age.

FIRST HALF		THEN	CHANCE A HOT STREAK STARTS
EXPLORE	→	NEVER NARROWED	↓ BELOW CHANCE
NEVER WANDERED	→	EXPLOIT	↓ BELOW CHANCE
EXPLORE	→	EXPLOIT	+20.5% ARTISTS +13.8% DIRECTORS +19.2% SCIENTISTS

Each half is read off the work itself — how spread out your styles or topics are, window by window: wide, then narrow. **Chance** is the same careers, shuffled. **Lift** is the gap against that.

Liu, Dehmamy, Chown, Giles & Wang (2021) *Nature Communications* 12:5392 · doi — 800,000 artworks, 79,000 films, 20,000 scientists; observational, no causal claim · no peak age — hot streaks land uniformly at random across a career; John Fenn's ended in a Nobel at 85: Liu et al. (2018) *Nature* 559:396–399 · doi

FIELD NOTES

Ten startups. Most didn't work.

BROKEN EGG GAMES

my first — volunteered

PROPHOUND

co-founded with postdocs

VEDA

acquired

58,111 startups — Yin, Wang, Evans & Wang (2019) *Nature* 575:190–194 · doi — eventual-successes and never-successes are indistinguishable on their first attempt; the winners discard less between attempts; clean-sheeting every time is the losing regime · the three ventures are from memory, not a registry — details as I remember them

Non-objective search has a prerequisite: margin.

> *"We would be remiss if the reader infers a message that novelty search is better than objective-based search."*
>
> Lehman & Stanley, 2011 — their own result

> *"If you're in a situation where you can't afford to take risks... just doing things because they're interesting could be pretty risky."*
>
> Ken Stanley, EP130

A visa tied to a contract. A dependent. Savings measured in weeks.

Lehman & Stanley (2011) "Abandoning Objectives," *Evol. Comput.* 19(2):189–223 · doi — a result about a population of hundreds, most discarded; you are one lineage · EP130, full transcript · an argument for building margin into a community — not for telling people to be braver

VERIFIED 7 AUGUST

One opens the day after this conference starts.

ROUTE	WHAT IT IS
ARIA · Nature-Inspired Computation	Programme Director call **opens 17 Aug**
Astera Residency	\$125–250k + up to \$1.5m — **no university required**
SFI Complexity Postdoc	any discipline · **closes 30 Sep**
JSPS Postdoc (Japan)	¥362,000/mo — **the host applies**, so ask

Also: MSCA Postdoctoral, 9.6% success, mobility rule applies; ERC relocation top-up to €2m for PIs moving from non-associated third countries · aria.org.uk · astera.org/residency · santafe.edu · jsps.go.jp · erc.europa.eu · ARIA eligibility for non-UK applicants unverified — check before you promise anyone anything

BACKUP

The rest of the argument.

Cut for time, not for doubt. Every one of these answers a question somebody in this room is about to ask.

Press **g**, a number, **Enter** to jump.

BACKUP · THE DYL EVIDENCE BASE, IF ANYONE ASKS

1,000,000+	0
READERS	CONTROLLED TRIALS

Zero RCTs, zero quasi-experimental studies, zero pre-post evaluations of the DYL curriculum. The academic "life design" literature (Savickas et al. 2009, *J. Vocational Behavior* 75(3) · doi) is a different framework sharing the name · career interventions do work modestly: Whiston et al. (2017) · doi · design-thinking critique: Micheli et al. (2019) · doi

BACKUP · ASKED AT EVERY ALIFE TALK: "ISN'T THE LANDSCAPE METAPHOR WRONG?"

- **Dimensionality — a real hit.** In high dimensions the good regions are *connected*: *"no need to cross any adaptive valleys."* A 1-D picture overstates being *trapped*.
- **No fixed surface** once fitness depends on everyone else — which is the whole crowding term.
- **"Career fitness" isn't a scalar.** One noisy sample a year.

Take the concession — it doesn't save you. A greedy walk halts after a mean of **$e - 1 \approx 1.72$ steps**, however big the space is.

You are never trapped. You just stop improving after about two moves.

Orr (2003) *J. Theor. Biol.* 220:241–247 · doi · Gavrilets & Gravner (1997) *JTB* 184:51–64 · doi — holds above a percolation threshold, under binary fitness; it establishes connectivity, not reachability · Kaplan (2008) *Biol&Phil* 23(5) · doi

BACKUP · THE FULL US FUNDING TABLE

Proposed, enacted, and enjoined are three different things.

	PROPOSED (FY2026 REQUEST)	ACTUALLY ENACTED
NSF, total	−56.9%	−3.4%
NSF Research & Related Activities	—	**flat** in nominal dollars
NSF STEM Education	—	−20%
All federal R&D	−20.5%	−0.2% (nondefense −5.2%)
NIH	—	+0.9%

The damage is real and it landed **unevenly** — but it is not the number that was in the headlines.

GRFP 2025 detail: ~1,500 awards, 10% success overall, **under 5% in the life sciences**; the 2026 round made 2,500 awards from ~14,000 applicants. · CRS R48783 Table 1 · H.R. 6938, signed 23 Jan 2026 · NIH via P.L. 119-75, 3 Feb 2026 · NSF release, 13 Apr 2026

BACKUP · THE HONEST CAVEAT ABOUT THE FORMALISM, IF ANYONE INVOKES ACTIVE INFERENCE

> *Active inference gives you the vocabulary — interoceptive and exteroceptive streams inside one generative model.*
>
> *But notice what it doesn't give you. The free energy principle tells you how an agent minimises surprise given its priors. It has nothing whatsoever to say about whose priors those are.*
>
> *Career unhappiness isn't an inference problem. It's a* ***prior*** *problem.*

Seth (2013) *TiCS* · doi · Seth & Friston (2016) *Phil Trans R Soc B* · doi · Friston conceded in 2018 that the principle cannot be falsified

BACKUP · STANLEY PRE-EMPTS THE NAIVE CAREER READING HIMSELF

> *"I'm not saying like, 'Okay, well now I want to figure out how to get into grad school. So I'll just wander around and do interesting things and hope that it will happen.'"*
>
> Ken Stanley, The Jim Rutt Show EP130

He is not against near-horizon goals.
He is against treating **a distant destination as a gradient.**

What the ALife toolkit actually is

METHOD	WHO IS PAYING FOR IT
Agent-based modelling	policy, epidemics, markets, logistics, crowd safety
Evolutionary computation, quality-diversity	design optimisation where no gradient exists
Cellular automata, self-organisation	smart materials, morphogenetic engineering
Multi-agent and collective behaviour	agentic AI orchestration, swarms, market design
Simulating counterfactual worlds	world models, digital twins, synthetic environments
Evaluating systems with no ground truth	the bottleneck problem in AI right now

ALife is the only community that has spent thirty-five years asking *how do you tell if something interesting happened when there's no score?*

Each right-hand cell is sourced elsewhere in this deck: the live-postings audit, 2026-08-07 (slides 21 & 23) · DeepMind & Sakana releases (slide 13) · the no-ground-truth bottleneck: the eval-team postings quoted verbatim on slide 23

The copying trap, corrected twice.

- **The left-hand arm isn't "not talking enough."** Lazer & Friedman's own density curve is a *fragmentation* artefact — their text says that among networks that stay in one piece, "the inverse-U shape disappears... the fewer connections the better in the long run." Sparse groups do badly because they are *several small populations drawing from a smaller pool*, not because they are quiet. That is why the live slide cites Derex instead.
- **The whole result is conditional on copying the best performer.** It reverses when people copy the most common answer instead.

The argument survives both: not one big room — many overlapping small ones.

Lazer & Friedman (2007) *ASQ* 52(4):667–694 · doi · Barkoczi & Galesic (2016) *Nature Communications* 7:13109 · doi

BACKUP · BERNSTEIN IN FULL, IF YOU WANT THE TABLE

You don't have to choose. Just stop talking sometimes.

Triads solving a rugged problem. Neighbours' solutions visible **every round**, **every third round**, or **never**.

INTERACTION	FOUND THE OPTIMUM	MEAN QUALITY
Constant	33.3%	best tier
Intermittent	**48.3%**	best tier
Isolated	44.1%	worst

Intermittent groups got **both halves** — they alternately converged and diverged. "Solutions improved most on rounds with social influence *after a period of separation*."

Bernstein, Shore & Lazer (2018) "How intermittent breaks in interaction improve collective intelligence," *PNAS* 115(35):8734–8739 · doi · PMC6126746 — the same Lazer, eleven years later

BACKUP · AND IT GETS WORSE FOR A ROOM THIS SIZE

Conformity rises with group size and with task difficulty.

699 participants. Challenging tasks elicit more copying, and the rate of copying increases with the number of people in the room.

A 550-person Discord is structurally at risk of converging on the same wrong answer — precisely on the questions that matter most.

Toyokawa, Whalen & Laland (2019) "Social learning strategies regulate the wisdom and madness of interactive crowds," *Nature Human Behaviour* 3:183–193 · doi.org/10.1038/s41562-018-0518-x

It is contested — and the critics wrote the paper *with* the advocate.

SCEPTICS	ADVOCATE
It's a category error: a *cause* of change, not a *process* of change.	The implications for evolutionary theory are profound.
No systematic direction — it raises and lowers fitness alike.	Ecological inheritance is a genuine second inheritance system.
It yields no unambiguous prediction.	"Not yet, but this is feasible."

And applied to a Discord server, it is frankly a metaphor. But protected niches in innovation studies, and the sociology of scientific movements, are not.

Scott-Phillips, Laland, Shuker, Dickins & West (2014) *Evolution* 68(5):1231–1243, open access · doi · Frickel & Gross (2005) *ASR* 70(2) · doi · Smith & Raven (2012) *Research Policy* 41(6) · pdf

BACKUP · MORE FROM THE HOT-STREAK DATA

Three things nobody expects

- **Productivity does not change during a hot streak.** Same output rate. Better output.
- People with a *lower* baseline benefit more.
- And the topic you end up exploiting is not the most recent, not the most cited, not the most popular one you tried.

Which is the same claim as Picbreeder, arriving from bibliometrics: you cannot tell which stepping stone matters while you are standing on it.

Liu et al. (2018) *Nature* 559:396–399 · doi · preprint arXiv:1712.01804

BACKUP · THE PIVOT RULE ALREADY EXISTS AS CODE

Intelligent Adaptive Curiosity, 2007.

It rewards **learning progress** — the rate at which prediction error is falling — not the error, and not any external reward.

The agent automatically abandons both what it has mastered and what it cannot learn, and self-organises developmental stages.

It moves on when a region stops teaching it anything. That is the pivot rule, implemented, nineteen years ago.

Oudeyer, Kaplan & Hafner (2007) *IEEE Trans. Evol. Comput.* 11(2):265–286 · inria.hal.science/hal-00793610

BACKUP · AND THE NIH NATURAL EXPERIMENT ON FAILURE

Near-misses outperformed narrow winners.

Researchers who just missed the funding line and persisted went on to produce **21% more hit papers** than those who just cleared it.

The setback also made 12.6% of them disappear from the record permanently.

Both halves are the finding. Say both.

Wang, Jones & Wang (2019) *Nature Communications* 10:4331 · PMC6773762

BACKUP · THE STANDARD ADVICE VS THE DATA

Seven pieces of advice, contradicted.

- *"Focus. Pick a lane."* → exploitation without prior exploration does worse than chance.
- *"Early wins compound."* → near-misses beat narrow winners by 21%.
- *"Opportunity is rare, hold on."* → the richer the environment, the sooner you leave.
- *"Explore your options" (the 37% rule)* → with cardinal payoffs the optimum is $\sqrt{n}$, not n/e. For 100 options, **10, not 37**.
- *"Set clear goals."* → objective search solved the hard maze 3/40; novelty search 39/40, with simpler solutions.
- *"Pivot decisively, clean sheet."* → reuse is what separates success from stagnation.
- *"Don't do anything rash."* → coin-flip-induced changers were happier six months later.

Bearden 2006 · doi · Lehman & Stanley 2011 · doi · Levitt 2021 · NBER w22487

BACKUP · THE NETWORKING RESULT IN FULL

The grubby feeling is measured.

INSTRUMENTAL NETWORKING → FEELS MORALLY DIRTY → NETWORKS LESS → PERFORMS WORSE

Build something. Let the network be a by-product.

Casciaro, Gino & Kouchaki (2014) "The Contaminating Effects of Building Instrumental Ties," *ASQ* 59(4):705–735 · doi — 165 lawyers: seniority predicted feeling less dirty ($b = -.25$, $p < .001$) · Corrigendum, ASQ 2024: the causal power-moderation interaction moved to $p = .072$ — worse-with-less-power is suggestive, not settled. Say so if asked.

BACKUP · ALIFE.ORG/JOBS — IF ANYONE ASKS "IS IT REALLY THAT EMPTY?"

I have never had a job titled "artificial life researcher."

Faculty | Industry | Postdoctoral | PhD studentship | Undergraduate internship

No items found.

The ISAL jobs board · alife.org/jobs · verified 2026-08-06 — re-check before Q&A if you plan to invoke it

BACKUP · MY OWN THESIS RESULT

A career with fixed rules will, provably, cycle.

Adams, A. M., Zenil, H., Davies, P. C. W. & Walker, S. I. (2017) "Formal Definitions of Unbounded Evolution and Innovation Reveal Universal Mechanisms for Open-Ended Evolution in Dynamical Systems," *Scientific Reports* 7:997 · doi.org/10.1038/s41598-017-00810-8 — only state-dependent rules keep innovating; fixed rules recur. Applied to a career it is a metaphor on purpose — concede that in the same breath.